

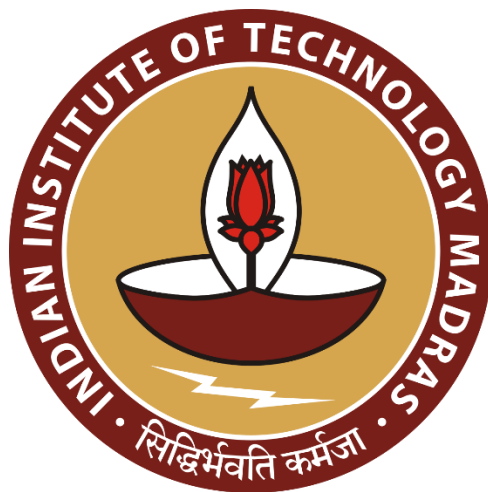
Multimodal Misinformation Detector Agent

Data Science and AI Lab Project

Milestone 4 Report

Group 9

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1. Objective

This milestone focuses on the initial training, fine-tuning, and experimentation of models for multimodal misinformation detection. The objective is to train and evaluate both text-based and vision-based classifiers that determine whether an input (text or image) is real or manipulated.

The milestone extends from previous work on data preprocessing and architecture design (Milestone 3), integrating model training, hyperparameter tuning, and performance evaluation.

2. Dataset Details

2.1 Image Dataset

- **Source** - [Kaggle – AI-Generated Images vs Real Images](#)
- **Structure:** Binary labelled dataset (*real* and *fake*) divided into train and test folders.
- **Sample count:** 48,000 train, 12,000 test images.
- **Preprocessing:**
 - Converted to RGB and resized to 224×224 px.
 - Normalized using CLIP's built-in vision processor
 - Shuffled and batched with PyTorch DataLoader.

2.2 Text Dataset

- **Source:** [HuggingFace - FEVER dataset \(Fact Extraction and VERification\)](#)
- **Structure:** Each entry contains a *claim*, *label*, and extracted *evidence sentences* from Wikipedia.
- **Sample count:** 20,000 balanced samples after filtering and sampling equal counts for *SUPPORTS* and *REFUTES*.
- **Preprocessing:**
 - Removed instances labeled *NOT ENOUGH INFO..*

- Filtered and chose claims with 1–10 evidence sentences.
- Sampled 10,000 examples per label for balanced training.
- Tokenization using microsoft/deberta-v3-base tokenizer (max length = 256).

3. Model Architecture

3.1 Image Model

- **Base Model** - *openai/clip-vit-base-patch32*
- **Approach:** Used CLIP's vision encoder with a custom classifier head:
 - Linear (512 → 256)
 - ReLU + Dropout(0.5)
 - Linear(256 → 1) + Sigmoid
- **Parameters:** CLIP frozen; only classification head trained.
 - Epochs = 5
 - Batch_size = 32
- **Loss Function:** Binary cross-entropy.
- **Optimizer:** Adam (LR = 1e-4).
- **Output:** Probability of fake image.

3.2 Text Model

- **Base Architecture:** *microsoft/deberta-v3-base* (Transformer encoder, disentangled attention).
- **Fine-Tuning:** Trained on FEVER dataset for 3 epochs.
- **Input:** Concatenated *claim* and *evidence sentences*.
- **Output: Binary label** — *Supports* (Real) or *Refutes* (Fake).
- **Loss Function:** Cross-entropy loss.
- **Evaluation Metrics:** Accuracy, F1-score.

Parameter	Value
Model	DeBERTa-v3-base
Epochs	3
Learning rate	2e-5
Batch size	16 (train) / 32 (eval)
Optimizer	AdamW
Weight decay	0.01
Mixed precision	FP16
Evaluation strategy	Per epoch

3.3 Explanation Generator

- **Model:** *mistralai/Mistral-7B-Instruct-v0.2*.
- **Function:** Generates a natural-language justification for the credibility decision, integrating both evidence retrieval and classification output.
- **Example Output:**
“The claim is likely false because multiple verified fact-check sources, including Google Fact Check and Wikipedia, contradict the provided statement.”

4. Training setup

- **Environment:** Kaggle Notebook with NVIDIA T4 GPU
- **Frameworks:** PyTorch, Hugging Face Transformers.
- **Loss Functions:** Cross-entropy (text), BCEWithLogitsLoss (image).
- **Metrics:** Accuracy, F1-score.
- **Training strategy:**
 - Gradient accumulation with early stopping.
 - Learning rate scheduling and weight decay regularization.
 - FP16 mixed-precision training for efficiency.

5. Regularization & Optimization Techniques

- **Dropout (0.5):** Used in image classifier head to prevent overfitting.
- **Weight Decay (0.01):** Improved stability in DeBERTa fine-tuning.
- **Early Stopping:** Avoided unnecessary training after convergence.
- **Data Shuffling & Balanced Sampling:** Prevented bias toward specific classes.

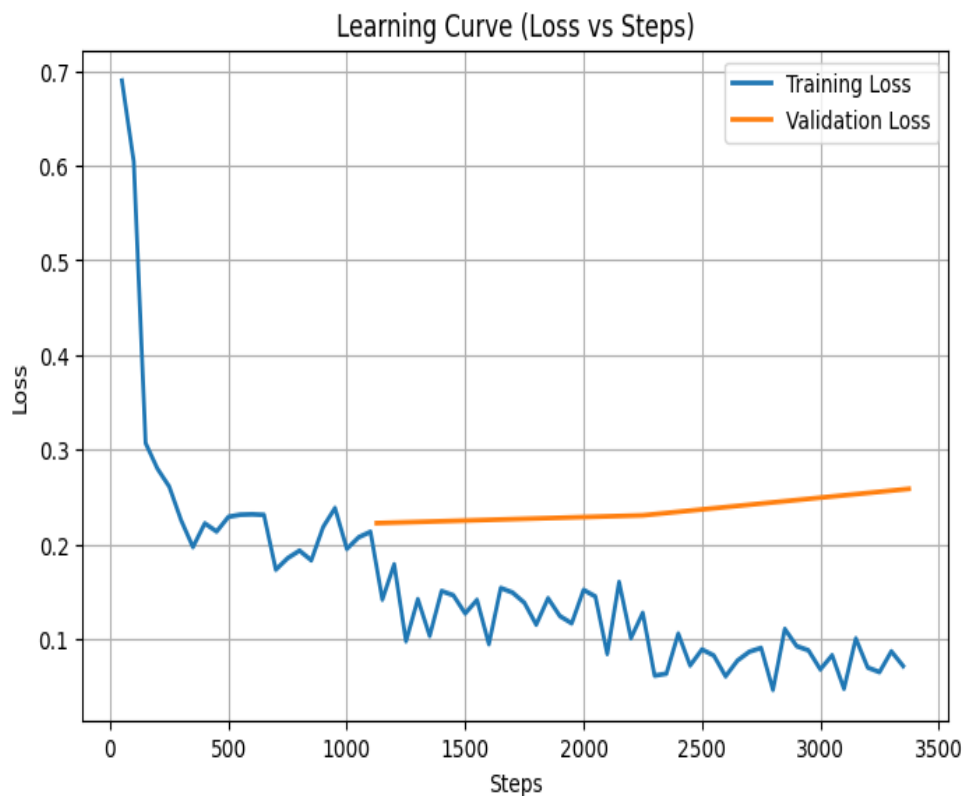
6. Initial Training Results

6.1 Text Model

Epoch	Training Loss	Validation Loss	Accuracy	F1
1	0.213000	0.222147	0.948000	0.948361
2	0.127300	0.230390	0.950000	0.950199
3	0.071100	0.258387	0.950000	0.950446

Final Evaluation Metrics:

```
eval_loss : 0.2584
eval_accuracy: 0.9500
eval_f1 : 0.9504
eval_runtime: 13.8990
eval_samples_per_second: 143.8950
eval_steps_per_second: 4.5330
epoch : 3.0000
```



6.2 Image Model

- Achieved ~0.91 validation accuracy after 4 epochs..
- Observed clear separation between real and AI-generated images.
- Model stable with dropout regularization, avoiding overfitting despite limited dataset size.

7. Model Artifacts

- **Text Model Checkpoints:** [*fever finetuned deberta/*](#)
- **Image Model Checkpoints:** [*clip finetuned classifier.pth*](#)
- **Tokenizer and Processor:** Hugging Face pre-trained modules.
- **Training Logs:** Generated via Hugging Face Trainer logs and PyTorch loss curves.
- **Reproducibility:** Training code pushed into [GitHub repository](#).

8. Observations & Notes

- Fine-tuning DeBERTa-v3 significantly improved accuracy to 95%.
- The image model showed robust early results despite limited tuning.
- Evidence retrieval accuracy influences final classification quality — planned enhancement through improved query formulation and document ranking.
- Mistral explanations align well with model outputs but need refinement.
- Next milestone (Evaluation & Analysis) will focus on cross-modal consistency checks, qualitative evaluation, and model calibration.

9. Summary

This milestone successfully established and validated the training phase of a multimodal misinformation detection system. Both the text-based classifier and image authenticity detector achieved high accuracy and generalization.